

Medical Image Data Analysis

: 3D MRI Generation via Flow Matching Model

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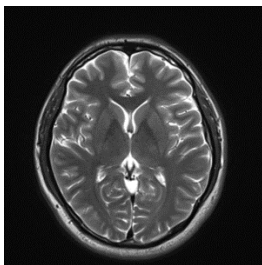


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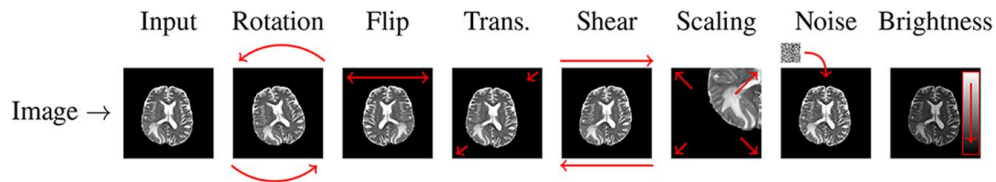
We Need More Medical Imaging Data

- Medical video/image is...
 - Hard to collect at scale.
 - Labeling costs a lot.
- Aggressive data augmentation/transformation could harm semantics.
 - Hard to collect at scale.

Label: ???

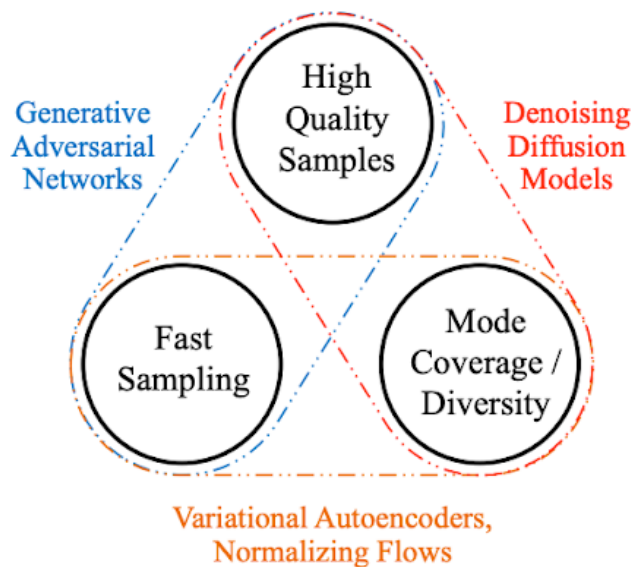


Dangerous



Medical Image Generation

- Diffusion & Flow matching model could help
 - Can do semantic preserved generation with conditional training/sampling.

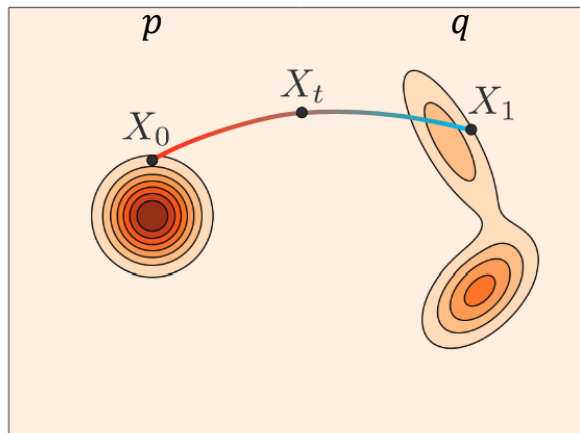


What is Flow Matching?

- **GOAL:** Build a flow ψ_t that transforms a source sample $X_0 \sim p$ into a target sample $X_1 \sim q$ where $0 \leq t \leq 1$.



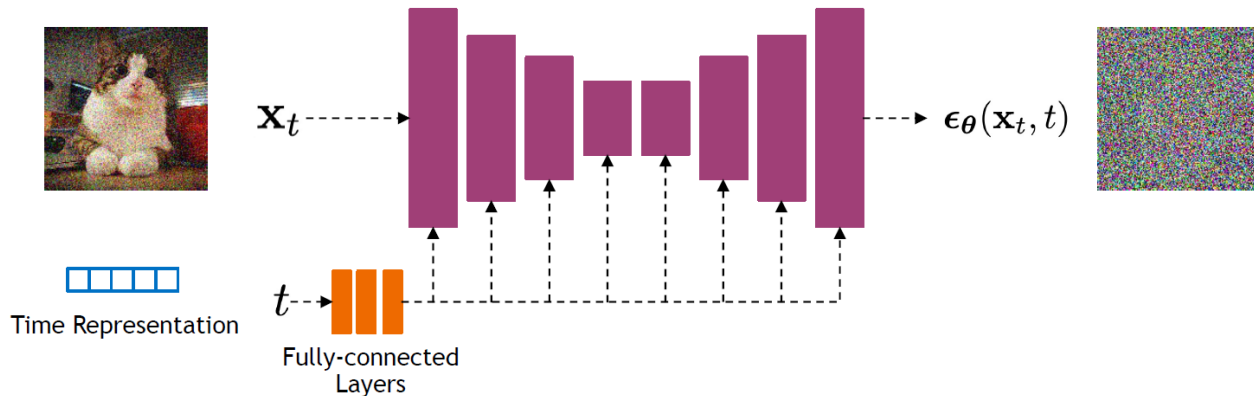
$$X_t = tX_1 + (1 - t)X_0$$



Image=Noise=Velocity Prediction Network

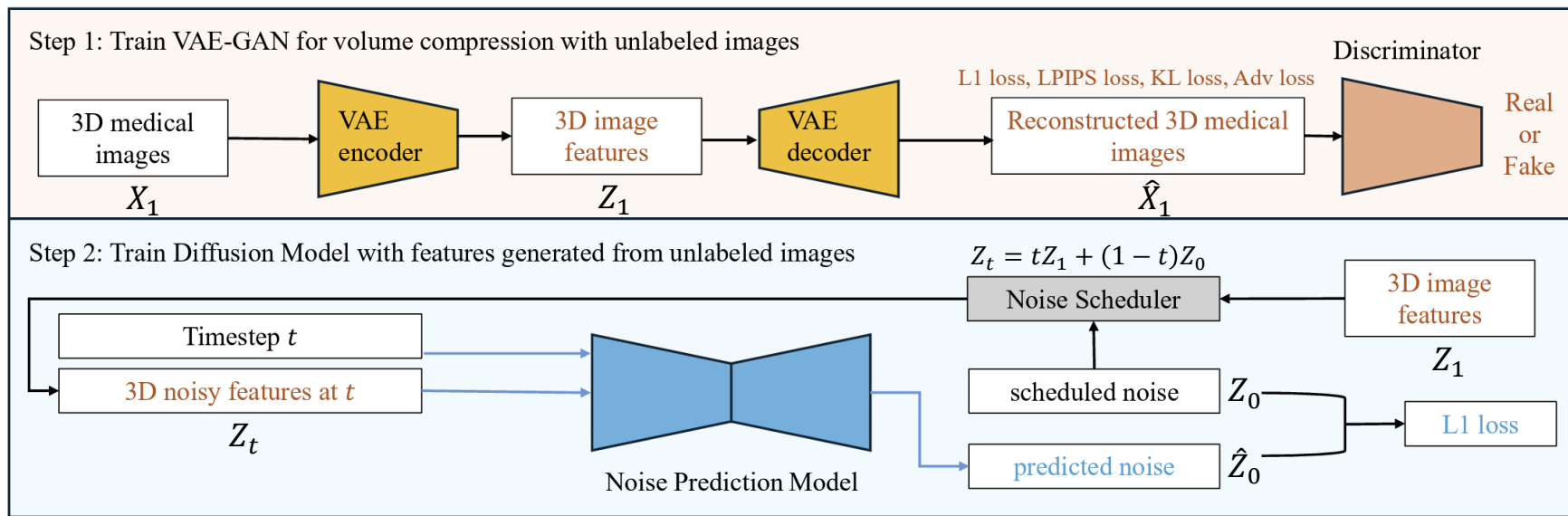
- Use U-Net architectures with ResNet blocks and self-attention layers to construct $\epsilon_{\theta}(\mathbf{x}_t, t)$.
 - Input: noisy image x_t and corresponding time point t .
 - Output: noise x_0 applied to input noisy image x_t . (=noise prediction)

$$X_t = tX_1 + (1 - t)X_0$$



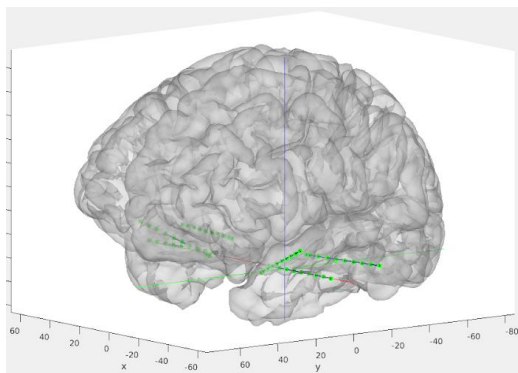
Latent Model Architecture

- **GOAL:** lower computational demands of training high-resolution image.
- **Solution:** explicitly separate image compression stage & generative learning stage.



BraTS Dataset

- Brain Tumor Segmentation Dataset
 - 3D spatial MRI
 - (1x240x240x155) raw matrix each
 - Grayscale channel
 - Modality: T1, T1-c, T2, Flair and mask.



Training

Glioma: 1251

Metastasis: 238

FLAIR	✓
T1	✓
T1-c	✓
T2	✓
mask	✓

Four modalities and seg.
masks are available

Validation

Glioma: 219

Metastasis: 31

FLAIR	✓
T1	✗
T1-c	✓
T2	✓
mask	✗

One random modality is
dropped for each case

BraTS Training

VAE

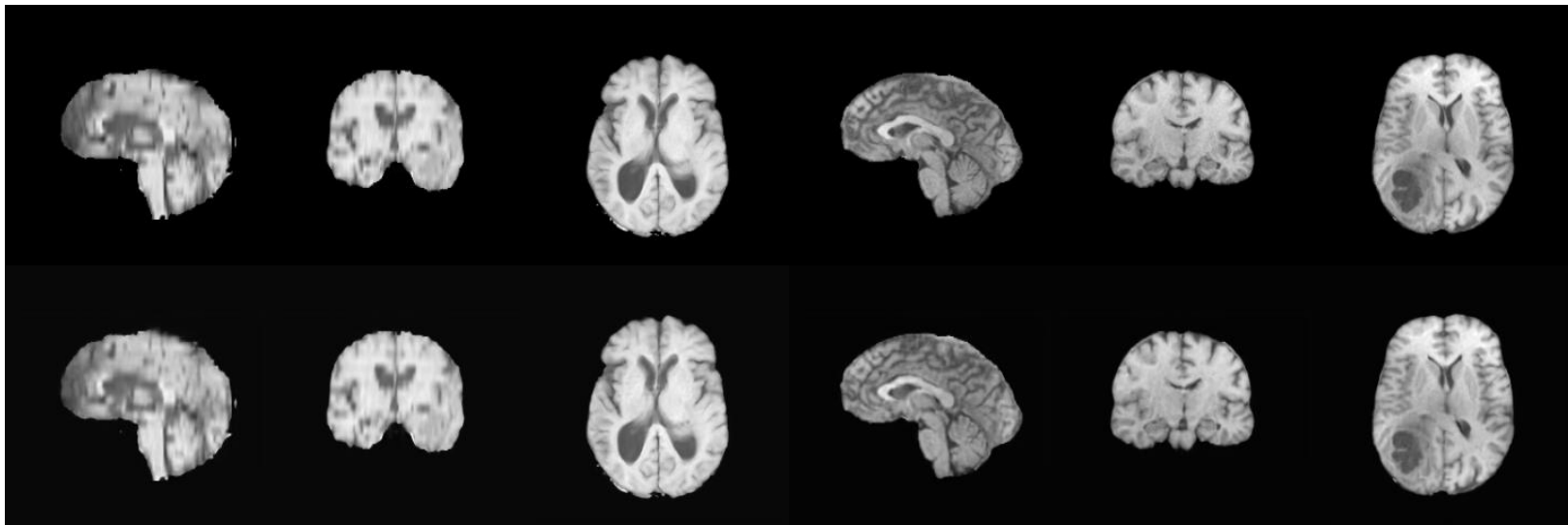


UNET



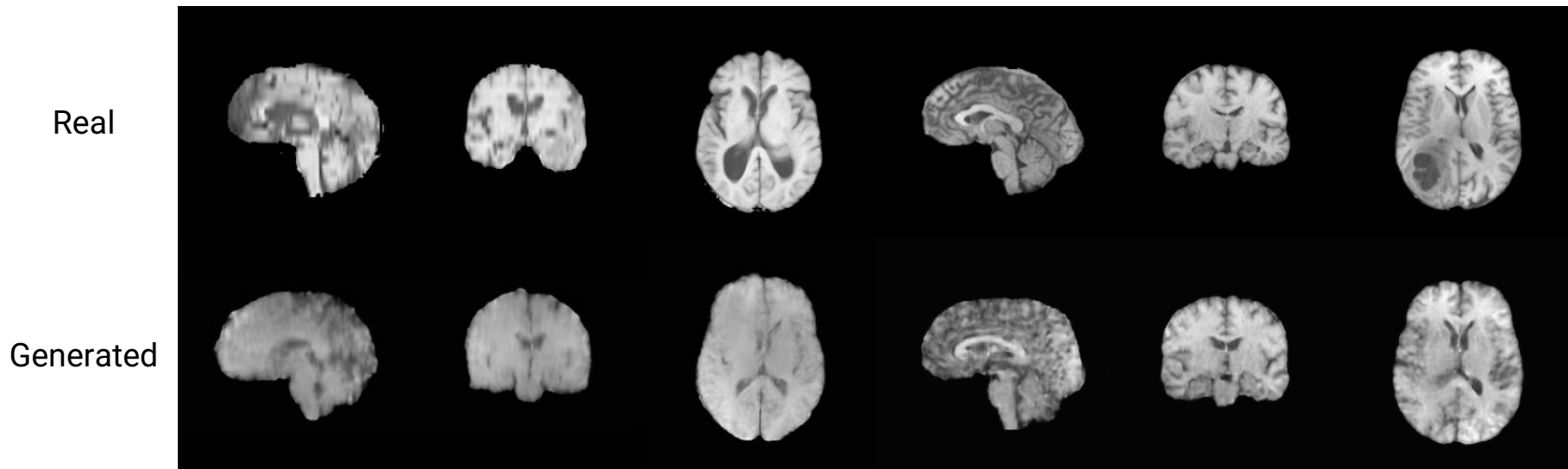
BraTS (VAE Reconstruction)

Real
(input)



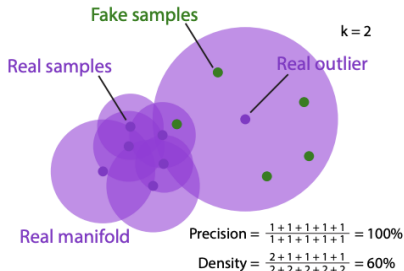
Recon
(output)

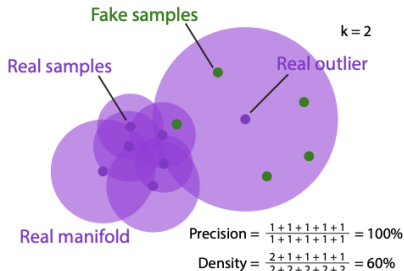
BraTS (Unet Generation)



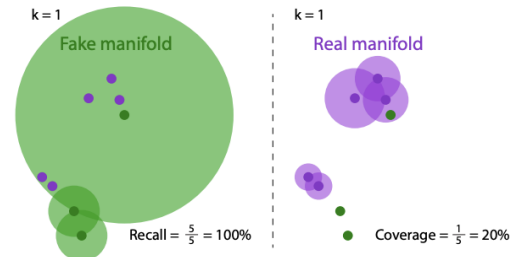
BraTS (Evaluation)

- Applied bootstrapping for sampling. (200 vs. 200 / 20 times)

BraTS (T1n)	Real vs. Recon	Real vs. Real (other)	Real vs. Gen (ckpt-100K)
FID ↓	0.0000 ± 0.0000	0.0021 ± 0.0003	0.0006 ± 0.0001
Precision ↑	0.9734 ± 0.0104	0.7421 ± 0.0260	0.7832 ± 0.0215
Recall ↑	0.9749 ± 0.0105	0.8193 ± 0.0168	0.5251 ± 0.0257
Density ↑	1.1717 ± 0.0326	0.8497 ± 0.0329	1.1223 ± 0.0382
Coverage ↑	0.9949 ± 0.0056	0.8015 ± 0.0213	0.7810 ± 0.0225
LPIPS ↓	0.019	 Precision = $\frac{1+1+1+1}{1+1+1+1} = 100\%$ Density = $\frac{2+1+1+1}{2+2+2+2} = 60\%$	
PSNR ↑	36.79		
SSIM ↑	0.9866		



(a) Precision versus density.



(b) Recall versus coverage.

Useful Resources

- Flow Matching Guide and Code: <https://arxiv.org/abs/2412.06264>
- MAISI paper: <https://arxiv.org/abs/2409.11169>
- High-Resolution Image Synthesis with Latent Diffusion Models:
<https://arxiv.org/pdf/2112.10752>
- BraTS website: <https://www.synapse.org/Synapse:syn64153130/wiki/630130>
- MONAI: <https://github.com/Project-MONAI/MONAI>
- MAISI tutorial: <https://github.com/Project-MONAI/tutorials/tree/main/generation/maisi>



Thank You!

